KAAIS23: Introduction to AI

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* Indicates required question

Email *

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The neurons in a neural network are connected:

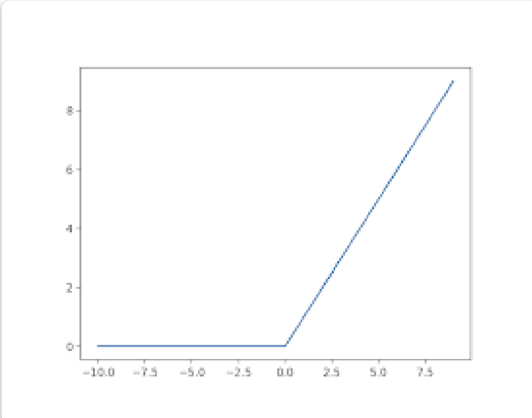
○ Each neuron of the neural network is connected to every other neuron of the neural network

○ Chaotic

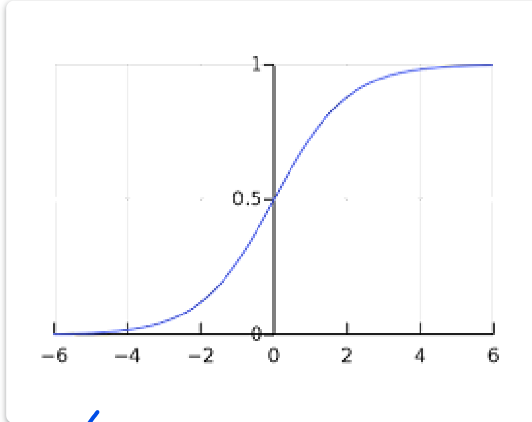
○ Each neuron of one layer of the neural network is connected with each neuron of the previous and next layers, there are no connections between neurons of one layer

○ All neurons of one layer are connected by edges

Which one of these plots represents a sigmoid activation function



a



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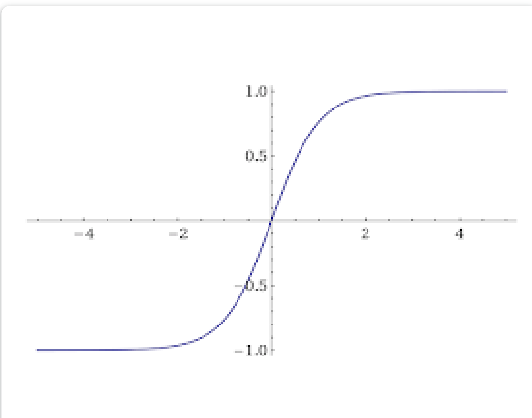
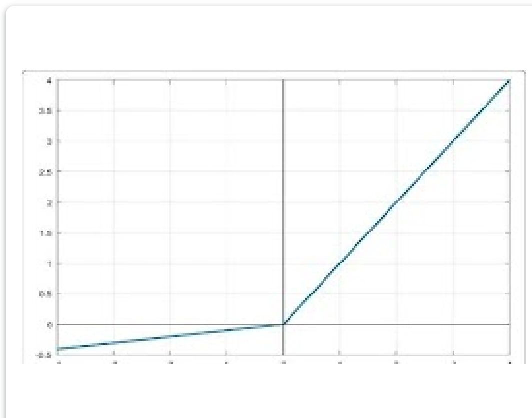


c



d

Select attributes that can only be considered categorical (and cannot be considered binary or real).

- [ ] Internet connection speed
- [ ] The bank client has a military ticket
- [ ] Year of birth
- [ ] Line of the mobile operator
- [ ] Type of house - brick, block, panel, etc.
- [ ] Car color

Finding the minimum of a function using the gradient descent method consists of iteratively moving in the direction of

- ○ the gradient of the function
- ○ None of the answers
- ○ of the antigradient (i.e., minus the gradient) of the function.

Please select your venue *

- ○ KAUST Group 1
- ○ KAUST Group 2
- ○ AlFaisal
- ○ KFUPM

The trainable parameters of the neural network are:

- ○ Neurons
- ○ Activation function
- ○ edge weights
- ○ All of the answers

What is the difference between logistic regression and linear regression?

- Logistic regression can handle multiple dependent variables, while linear regression can only handle one.
- Logistic regression uses the sigmoid function, while linear regression uses the step function.
- Logistic regression predicts probabilities, while linear regression predicts actual values.
- Logistic regression is used for regression problems, while linear regression is used for classification problems.

Assume a simple MLP model with 3 neurons and inputs= 1,2,3. The weights to the input features are 4,5 and 6 respectively. Assume the activation function is f(x) = 3*x and the bias term is zero. What will be the output ?

- 96
- 32
- 64
- 128

Gradient decent algorithm is guaranteed to provide the global minima in all cases

- True
- False

Select the correct statements about regularization of neural networks.

- Increasing the regularization factor reduces overfitting of the network.
- Increasing the regularization factor increases the accuracy of approximation.
- Regularization loads the tuning of parameters.
- Regularization reduces the number of parameters of a neural network.

If the second derivative is positive, then the function

- has an extremum at this point.
- increases.
- is decreasing.
- convex.
- concave.

Overfitting is a phenomenon in which an algorithm obtained during training is...

- shows the same quality on the training sample and the new data.
- shows lower quality on the training sample than on the new data.
- shows lower quality on the new data than on the training sample.

Which hyperparameter would you change and how, if the training loss of a neural network is always increasing with epochs. (Assuming there are no other errors in the code)

- ◯ Decrease the learning rate
- ◯ None of the answers
- ◯ Increase the number training epochs
- ◯ Decrease the number of training epochs

Which of the following numpy library functions can be used to calculate the determinant of a matrix?

- ◯ np.linalg.inv(...)
- ◯ np.linalg.det(...)
- ◯ np.det(...)
- ◯ np.inv(...)
- ◯ np.transpose(...)

Learning rate, number of iterations, and the size of the hidden layers in neural networks are considered to be hyperparameters.

- ◯ False
- ◯ True

The first layer is called the?

- ◯ hidden layer
- ◯ input layer
- ◯ None of the above
- ◯ outer layer

Name *

Your answer

Let's consider the feature "Number of client's calls to the bank's support service". It accepts only non-negative integer values. What type does this attribute have?

- ◯ Binary - why not?
- ◯ Real - because integers are also real.
- ◯ Categorical - after all, it can only accept values from a finite set.

Cross-validation is used for

- ◯ Machine learning algorithm retraining on additional data
- ◯ improving the quality of the algorithm
- ◯ more accurate evaluation of the algorithm's performance
- ◯ interpretation of machine learning model features

Which of these are reasons for Deep Learning recently taking off?

- [x] We have access to a lot more computational power
- [x] Deep learning has resulted in significant improvements in important applications such as online advertising, speech recognition, and image recognition
- [x] We have access to a lot more data
- [x] Neural Networks are a brand new field

In a simple MLP model with 8 features in the input layer, 5 neurons in the hidden layer and 1 neuron in the output layer. What is the size of the weight matrices between the input and hidden layer, and between the hidden and output layer (note that z = w*x + b)?

- (x) [5 x 1] , [8 X 5]
- ( ) [5 X 8], [1 X 5]
- ( ) [8 X 5] , [5 X 1]
- ( ) [8 X 5] , [1 X 5]

Choose the correct statements about the derivative of the function.

- [x] If a function has an extremum at a point, and the derivative of the function is defined in it, then at this point the derivative of the function is zero
- [ ] If a function is continuous at a point, at that point the function has a derivative
- [x] If the derivative of a function at a point is zero, at that point the function has an extremum
- [ ] None of the answers
- [x] If a function has a derivative at a point, then the function is continuous at that point

Select the correct statements about neural network methods. ?

- [ ] Using a multilayer neural network, a high accuracy approximation can be obtained.
- [ ] Class labels in two-class classification can take values from the sets of both {0,1} and {-1,1}.
- [ ] A neural network that correctly approximates the training sample and poorly approximates the control sample is undertrained.

The output for the Python code below is
[val for val in np.arange(10) if val%5 ==0]

- ( ) [0,5]
- ( ) None of the above
- ( ) [0,1,2,3,4,5,6,7,8,9]
- ( ) [0,5,10]

In the process of training a neural network, objects are fed to the network as input:

- ( ) In batches with an arbitrary number of objects.
- ( ) In batches of 32 objects
- ( ) Strictly one at a time
- ( ) The whole dataset at once

What will be the shape of x in the code below

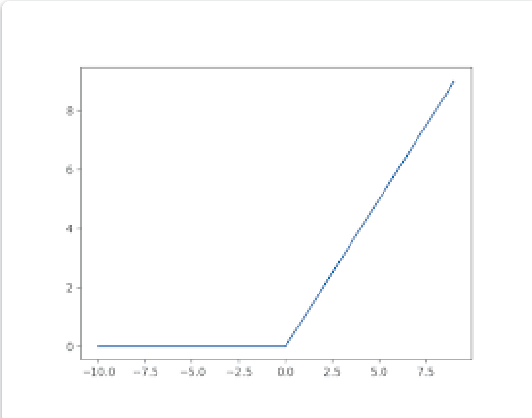
```
x = np.random.rand(3,4,5)
x = x.reshape(-1,1)
```

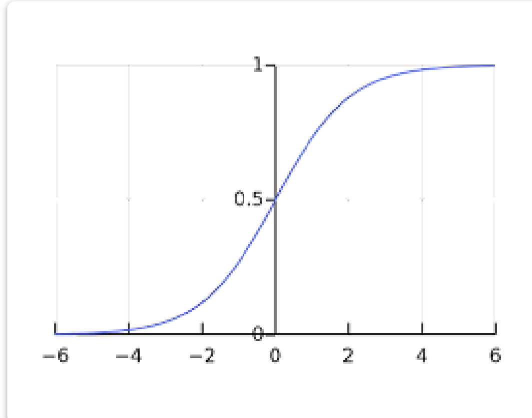
- ◯ None of the above
- ◯ (60,1)
- ◯ (1,60)
- ◯ (3,4,5)

How should the gradient descent step behave? Select all the correct observations.

- ☐ Take a sign depending on the value and direction of the gradient.
- ☐ Be small at first, then increase.
- ☐ Decrease as it approaches an extremum, or be constant but small.
- ☐ Be positive.

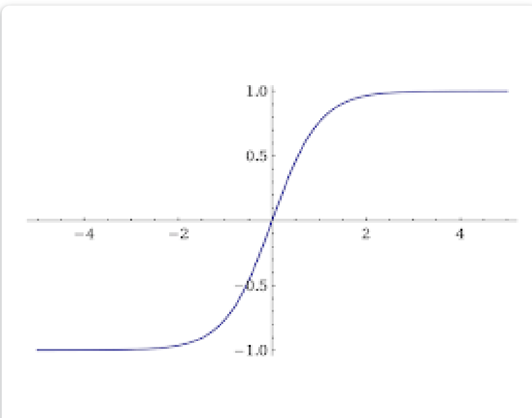
Which one of these plots represents a ReLU activation function

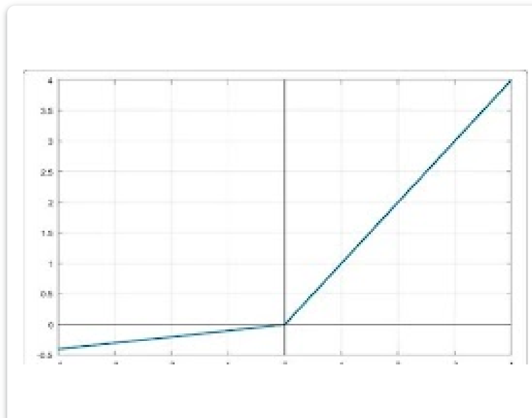




○ a

○ b





○ c

○ d

The derivative of a composition of functions f(g(x)) is calculated as the

☐ sum of derivatives of functions f and g on their arguments.

☐ product of the derivatives of functions f and g by their arguments.

We usually use cross entropy loss for classification and for linear regression problems.

- [x] False
- [ ] True

What is the sigmoid function used for in logistic regression?

- [ ] To normalize the input features to have zero mean and unit variance.
- [x] To calculate the predicted probability.
- [ ] To map the input features to a higher-dimensional space.
- [ ] To regularize the model coefficients and prevent overfitting.

Which neural network has only one hidden layer between the input and output?

- [x] Shallow neural network
- [ ] Deep neural network
- [ ] Feed-forward neural networks
- [ ] Recurrent neural networks

How does Ridge regression differ from Lasso regression in linear regression?

- ○ Ridge adds the absolute values of coefficients to the cost function, while Lasso adds the squared values.
- ○ Ridge adds the squared values of coefficients to the cost function, while Lasso adds the absolute values.
- ○ Ridge includes all features in the model, while Lasso selects a subset of features.
- ○ Ridge and Lasso are identical in linear regression.

Which of the following error functions is used in regression problems and is differentiable?

$$Q(w) = \sum_{i=1}^{\ell} |a(x_i, w) - y_i| \quad (1)$$

$$Q(w) = \sum_{i=1}^{\ell} [sign(a(x_i, w)) = y_i] \quad (2)$$

Text

$$Q(w) = \sum_{i=1}^{\ell} (a(x_i, w) - y_i)^2 \quad (3)$$

$$Q(w) = -\sum_{i=1}^{\ell} \left(y_i \ln a(x_i, w) + (1 - y_i) \ln(1 - a(x_i, w))\right) \quad (4)$$

- ○ (1)
- ○ (2)
- ○ (3)
- ○ (4)

Select the correct statements about the methods of tuning the parameters of neural networks.

?

- ○ Gradient descent algorithms are always preferred over stochastic gradient descent algorithms.
- ○ Gradient descent algorithms require differentiable error functions.
- ○ The error function has a single minimum.
- ○ A stochastic gradient algorithm necessarily uses all elements of the sample.
- ○ The gradient descent algorithm will find a global minimum in any problem.

What is the activation function σ should be taken in a single-layer neural network to produce a logistic regression?

$$a(x, w) = \sigma\left(\sum_{j=1}^{d} w_j^{(1)} x_j + w_0^{(1)}\right)$$

- ○ None of the above
- ○ σ(u) = 1/(1+exp(-u))
- ○ σ(u)=sign(u)
- ○ σ(u)=u

The number of input features is 10 and we have one hidden layer with 5 neurons, and one output neuron. What is the number of learnable parameters of this NN?

- ○ 55
- ○ 50
- ○ 16
- ○ None of the answers

Select the correct statements about activation functions.

- ☐ Before obtaining a sample, we can specify the optimal type of activation function of the network neurons.
- ☐ The activation function is necessarily differentiable.
- ☐ The sigmoidal logistic activation function determines the probability that an object belongs to a class.
- ☐ An identical (linear) activation function can be used to solve a regression problem.

What is the activation function σ should be taken in a single-layer neural network to produce a linear regression?

$$a(x, w) = \sigma\left(\sum_{j=1}^{d} w_j^{(1)} x_j + w_0^{(1)}\right)$$

- ○ None of the answers
- ○ σ(u)=u
- ○ σ(u)=sign(u)
- ○ σ(u) = 1/(1+exp(-u))

Why do we need batches during training? (choose all that applies)

- [x] Batching data optimizes the use of resources while training.
- [ ] Using batches is not conventional for training, and only used for testing.
- [x] Batches help train the model more efficiently.
- [ ] Using batches is applied for large datasets to optimize training.

Choose the correct statements about an arbitrary function f(x).

- [x] To each value of x from the domain of a function corresponds f(x) - the value of the function at a point.
- [ ] None of the answers
- [x] Only one argument value corresponds to each function value
- [x] For each real value of x is assigned some value of f(x)

National ID *

Your answer

In which of the following applications can we possibly use deep learning to solve the problem?

- ○ Protein structure prediction
- ✓ All of the answers
- ○ Detection of exotic particles
- ○ Prediction of chemical reactions

What does a neuron compute?

- A neuron computes a linear function (z = Wx + b) followed by an activation function
- A neuron computes an activation function followed by a linear function (z = Wx + b)
- A neuron computes the mean of all features before applying the output to an activation function
- A neuron computes a function g that scales the input x linearly (Wx + b)

Why do we initialize weights randomly instead of a fixed value (i.e. 0.5)

- The random initialization provides efficient learning, meanwhile, fixed initialization provides no learning.
- ✓ The random and fixed initializations are the same in efficiency and effectiveness, but random initialization is more conventional and recommended.
- The random initialization provides efficient learning, meanwhile, fixed initialization provides less efficient learning.
- The random initialization provides efficient learning, meanwhile, fixed initialization provides faster learning.

What is the difference between gradient descent and stochastic gradient descent?

- In stochastic gradient descent, normal noise is added to the antigradient at each iteration.
- In stochastic gradient descent, only one summand in the error functional is used at each iteration.
- None of the answers
- In stochastic gradient descent, a step in the random direction is taken at each iteration.

Values in the neurons of the output layer of the neural network for the classification task:

- Contains zeros and ones: a one in the neuron that corresponds to the class to which the neural network assigned the incoming object.
- Contains the probabilities of the object belonging to each of the classes.
- Contains arbitrary numbers: the lower the number, the higher the probability of the object belonging to the class corresponding to the neuron.
- Contains arbitrary numbers: the higher the number, the higher the probability of the object belonging to the class corresponding to the neuron.

A minima reached by the gradient decent algorithm for convex function is guaranteed to be a global minima

- False
- True

The fully connected layer of the neural network converts a vector of dimension 10 into a vector of dimension 5 and has the following parameters: weight matrix W and a vector of free terms
b. How many total numerical trainable parameters does such a layer have? In other words, compute the total number of variable weights in W and b.

Your answer

Find the derivative of the function
at the point x = 3.

$$f(x) = 2x^3 - x^2 + 1$$

Your answer

A vector whose coordinates are the partial derivatives of a function at a given point

- [ ] None of the answers
- [ ] Not called anything, this object is not useful at all
- [ ] It is called the gradient of the function at this point
- [ ] Specifies the direction of the fastest decreasing direction of the function at this point
- [ ] Sets the direction of the fastest increasing direction of the function at this point

Email *

Your answer

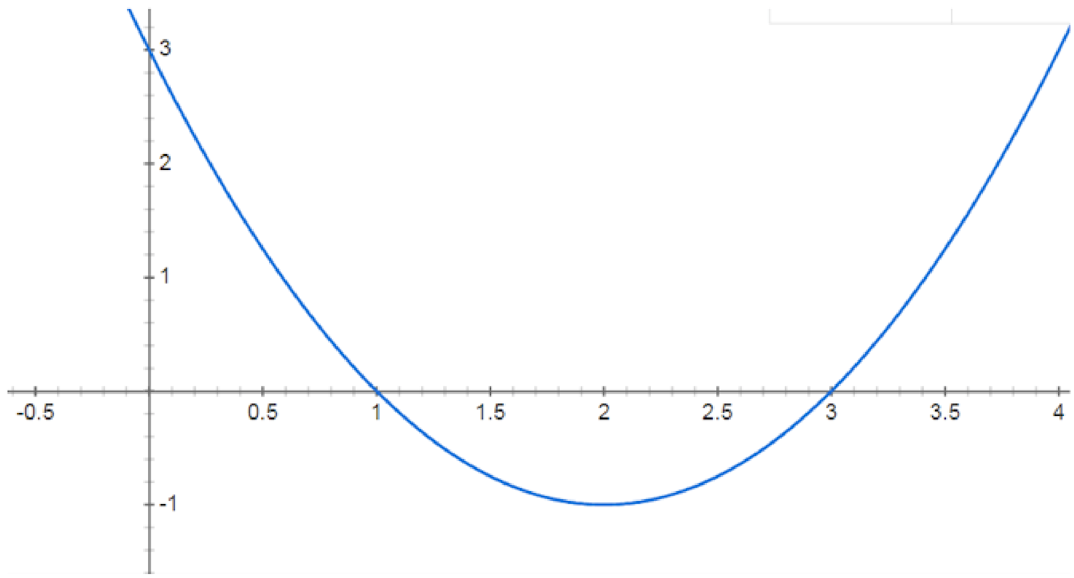
Overfitting happens when a model learns both data dependencies and random fluctuations,
meaning that the model learns the data too well.

- ◯ True
- ◯ False

For a neural network of three layers, if all layers are linear (i.e. activation function is Identity), what would happen to the neural network?

- [ ] Error, the network must have more than three layers.
- [ ] Error, the network must have non-linearity layers as well.
- [ ] No error, but the network will not learn.
- [ ] No error, the network will think of them as a single linear layer.

What can be said about the 1st and 2nd derivatives of a function?



- [ ] The 1st derivative is greater than 0 for x on the interval (-∞;2)
- [ ] The 2nd derivative is positive over the whole domain of definition
- [ ] The 1st derivative grows over the whole domain

Which of the following is a subset of machine learning?

- Pandas
- Numpy
- Deep Learning
- All of the answers

Suppose you have N input features per example, and that our data points are stored in X as X = [x1,x2,x3,...,xm]. What is the dimension of X?

- (m,m)
- (N,m)
- (N,N)
- (m,N)

You need to predict whether the dollar will rise or fall tomorrow. What kind of task is this?

- Classification
- Regression
- Clusterization

Underfitting occurs when a model can’t accurately capture the dependencies among data,
usually as a consequence of its own simplicity.

- False
- True

The two strong features of PyTorch are

- Nd arrays and matrix operations
- None of the above
- GPU support and Auto-differentiation
- CPU Support and GPU support

In logistic regression, what is the purpose of regularization?

- To prevent overfitting by penalizing large coefficients
- To speed up convergence
- To increase model complexity
- To reduce bias

If a function is strictly decreasing, it follows that the derivative.

- [ ] Equal to zero.
- [ ] Positive.
- [ ] None of the answers
- [ ] Negative.

You are given a set of 10,000 e-mails sent by one and the same person, and you need to group them in such a way that e-mails on similar topics - for example, personal correspondence, e-mails with airline tickets, etc. - are in the same group. What kind of task is this?

- ○ None of the answers
- ○ Clusterization
- ○ Regression
- ○ Classification

The last layer of nerual network for binary classification problems can have any activation function, e.g. ReLU, leak_relu, sigmoid etc

- ○ False
- ○ True

How many solutions can a system of linear equations have? (select all correct answers)

- [ ] None
- [ ] None of the answers
- [ ] Infinite number
- [ ] More than one, but a finite number
- [ ] One

You need to predict what the dollar exchange rate will be tomorrow. What kind of task is this?

- ○ None of the answers
- ○ Clusterization
- ○ Classification
- ○ Regression

Which of the following functions can be used as an activation function in the output layer if we wish to predict the probabilities of n classes (p1, p2..pn) such that the sum of p over all n equals to 1?

- ○ ReLu
- ○ Tanh
- ○ Sigmoid
- ○ Softmax

Which of the following statements about the cost function for logistic regression is true?

- ○ The cost function for logistic regression is always symmetric around the decision boundary.
- ○ The cost function for logistic regression is concave
- ○ The cost function for logistic regression is convex
- ○ The cost function for logistic regression penalizes large values of the model coefficients, which helps prevent overfitting.
- ○ The cost function for logistic regression is the same as the cost function for linear regression.
- ○ Other:

Select the correct statements.

- [ ] A function can have no more than two points of global extremum.
- [ ] A function can have at most one point of global extremum.
- [ ] Extrema is the minimum or maximum of a function.
- [ ] None of the answers
- [ ] The point at which the derivative of a function equals zero is called the extremum of the function.
- [ ] A function can have infinitely many points of global extremum.

What happens when the .backward() method of the loss function is called in Pytorch?

- The model weights are updated according to the calculated gradients
- Gradients are becoming zero
- Derivatives of the loss function are calculated using the trained parameters of the neural network
- The neural network output value is calculated

Which of the following error functions is used in classification tasks and is differentiable?

$$Q(w) = \sum_{i=1}^{\ell} |a(x_i, w) - y_i| \qquad \textbf{(1)}$$

$$Q(w) = \sum_{i=1}^{\ell} [sign(a(x_i, w)) = y_i] \qquad \textbf{(2)}$$

$$Q(w) = \sum_{i=1}^{\ell} (a(x_i, w) - y_i)^2 \qquad \textbf{(3)}$$

$$Q(w) = -\sum_{i=1}^{\ell} (y_i \ln a(x_i, w) + (1 - y_i) \ln(1 - a(x_i, w))) \qquad \textbf{(4)}$$

- (1)
- (2)
- (3)
- (4)

Assume the sigmoid function is denoted by z, what is the derivative of z

$$z = \frac{1}{1 + e^{-x}}$$

$$z - z^2$$

- [ ] A

$$z + z^2$$

- [ ] B

$$1 - z$$

- [ ] C

$$1 + z^2$$

- [ ] D

$$1 - z^2$$

- [ ] E

In the mean squared error function, our task is to find the value of the weigths for which the
loss function is the

- Gradient matrix
- Maximum
- Minimum
- Average

Matrices A and C have dimensions 2×7 and 9×6 respectively. It is known that there exists a product ABC. What is the size of the matrix B?

Your answer

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